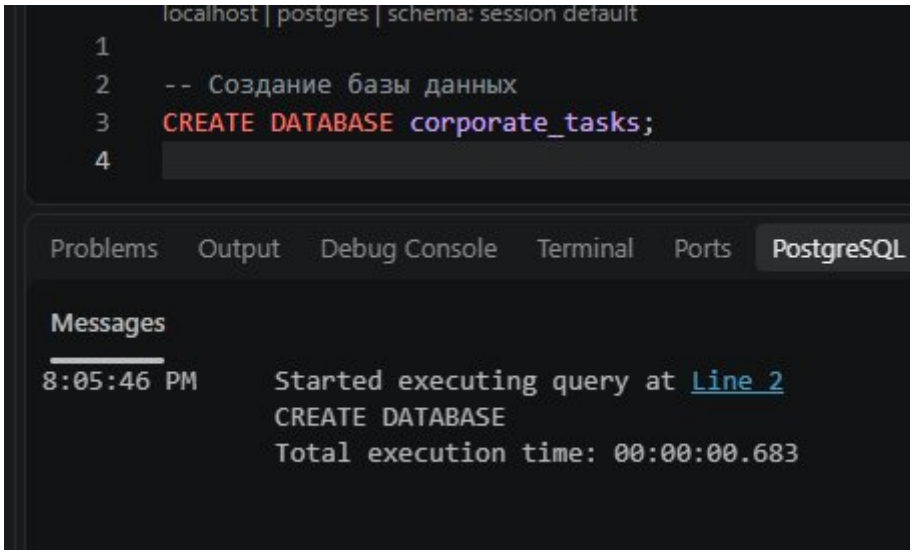
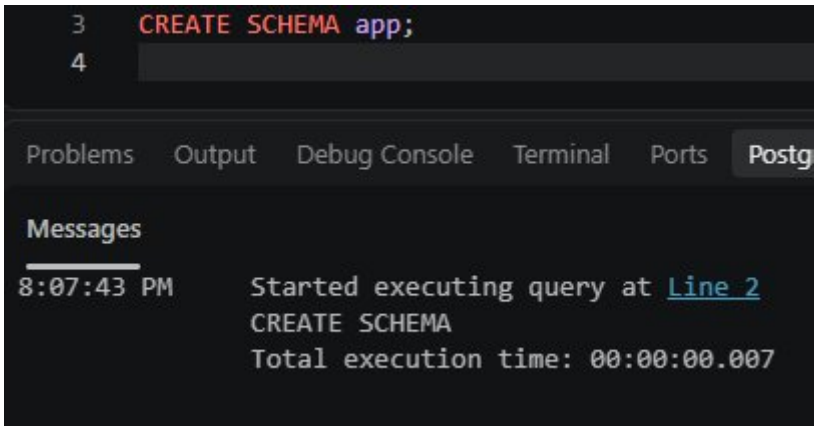
# 1.1




Создание базы данных



```sql
-- 1. Отделы
CREATE TABLE app.departments (
    department_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL UNIQUE,
    description TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

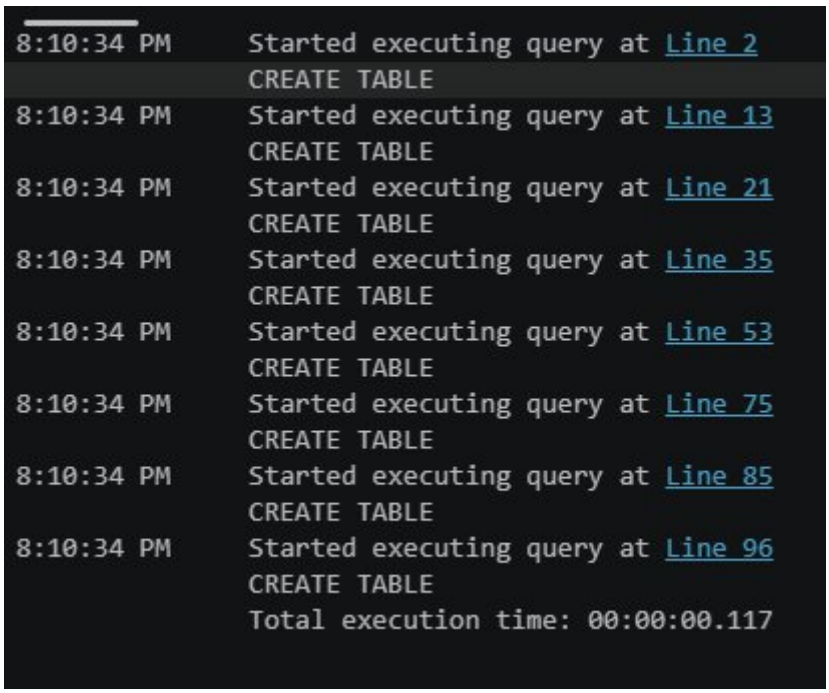
-- 2. Должности
CREATE TABLE app.positions (
    position_id SERIAL PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    level INTEGER NOT NULL CHECK (level BETWEEN 1 AND 10),
    description TEXT
);

-- 3. Пользователи
CREATE TABLE app.users (
    user_id SERIAL PRIMARY KEY,
    username VARCHAR(50) UNIQUE NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    full_name VARCHAR(100),
    department_id INTEGER REFERENCES app.departments(department_id),
    position_id INTEGER REFERENCES app.positions(position_id),
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 4. Проекты
CREATE TABLE app.projects (
    project_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    description TEXT,
    owner_id INTEGER REFERENCES app.users(user_id),
    department_id INTEGER REFERENCES app.departments(department_id),
    status VARCHAR(20) DEFAULT 'active'
        CHECK (status IN (
            'planning', 'active', 'on_hold',
            'completed', 'cancelled'
        )),
    budget NUMERIC(12, 2),
    start_date DATE,
    end_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Untitled-2

```sql
-- 5. Задачи
CREATE TABLE app.tasks (
    task_id SERIAL PRIMARY KEY,
    project_id INTEGER REFERENCES app.projects(project_id),
    title VARCHAR(200) NOT NULL,
    description TEXT,
    status VARCHAR(20) DEFAULT 'todo'
        CHECK (status IN (
            'todo', 'in_progress', 'review',
            'done', 'cancelled'
        )),
    priority INTEGER DEFAULT 3
        CHECK (priority BETWEEN 1 AND 5),
    assignee_id INTEGER REFERENCES app.users(user_id),
    created_by INTEGER REFERENCES app.users(user_id),
    estimated_hours NUMERIC(6, 2),
    actual_hours NUMERIC(6, 2),
    due_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
-- 6. Комментарии
CREATE TABLE app.comments (
    comment_id SERIAL PRIMARY KEY,
    task_id INTEGER REFERENCES app.tasks(task_id),
    user_id INTEGER REFERENCES app.users(user_id),
    content TEXT NOT NULL,
    is_internal BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
-- 7. История изменений
CREATE TABLE app.task_history (
    history_id SERIAL PRIMARY KEY,
    task_id INTEGER REFERENCES app.tasks(task_id),
    changed_by INTEGER REFERENCES app.users(user_id),
    changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    field_name VARCHAR(50) NOT NULL,
    old_value TEXT,
    new_value TEXT
);
-- 8. Журнал доступа
CREATE TABLE app.access_logs (
    log_id SERIAL PRIMARY KEY,
    user_id INTEGER REFERENCES app.users(user_id),
    action VARCHAR(50) NOT NULL,
    resource_type VARCHAR(50),
    resource_id INTEGER,
    ip_address INET,
    logged_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```




Созданы 8 таблиц

```sql
-- 1. Отделы
INSERT INTO app.departments (name, description)
VALUES
('IT Department', 'Information Technology'),
('Marketing', 'Marketing and Communications'),
('Sales', 'Sales Department');

-- 2. Должности
INSERT INTO app.positions (title, level)
VALUES
('Intern', 1),
('Junior Specialist', 2),
('Specialist', 3),
('Senior Specialist', 4),
('Manager', 5);

-- 3. Пользователи
INSERT INTO app.users
(username, email, password_hash, full_name,
 department_id, position_id)
VALUES
('alice', 'alice@company.com', 'demo_hash_alice',
 'Alice Johnson', 1, 4),
('bob', 'bob@company.com', 'demo_hash_bob',
 'Bob Smith', 1, 2),
('charlie', 'charlie@company.com', 'demo_hash_charlie',
 'Charlie Brown', 1, 3),
('diana', 'diana@company.com', 'demo_hash_diana',
 'Diana Prince', 2, 5),
('eve', 'eve@company.com', 'demo_hash_eve',
 'Eve Wilson', 2, 3);

-- 4. Проекты
INSERT INTO app.projects
(name, description, owner_id, department_id, status, budget)
VALUES
('Website Redesign', 'Redesign company website',
 1, 1, 'active', 50000),

('Mobile App', 'Develop mobile application',
 1, 1, 'planning', 100000);
```

```sql
-- 5. Задачи
INSERT INTO app.tasks
(project_id, title, status, priority, assignee_id, created_by)
VALUES
(1, 'Design homepage', 'in_progress', 2, 2, 1),
(1, 'Implement navigation', 'todo', 3, 3, 1),
(1, 'Setup CI/CD', 'todo', 2, 3, 1),
(1, 'Create color palette', 'review', 3, 2, 1),
(1, 'Prepare responsive layout', 'todo', 2, 2, 1),
(1, 'Test website', 'todo', 2, 3, 1),

(2, 'Requirements analysis', 'in_progress', 1, 1, 1),
(2, 'Architecture design', 'todo', 1, 3, 1),
(2, 'Setup project structure', 'todo', 2, 2, 1),
(2, 'Design login screen', 'todo', 3, 2, 1),
(2, 'Implement API', 'todo', 2, 3, 1),
(2, 'Test mobile app', 'todo', 2, 1, 1);

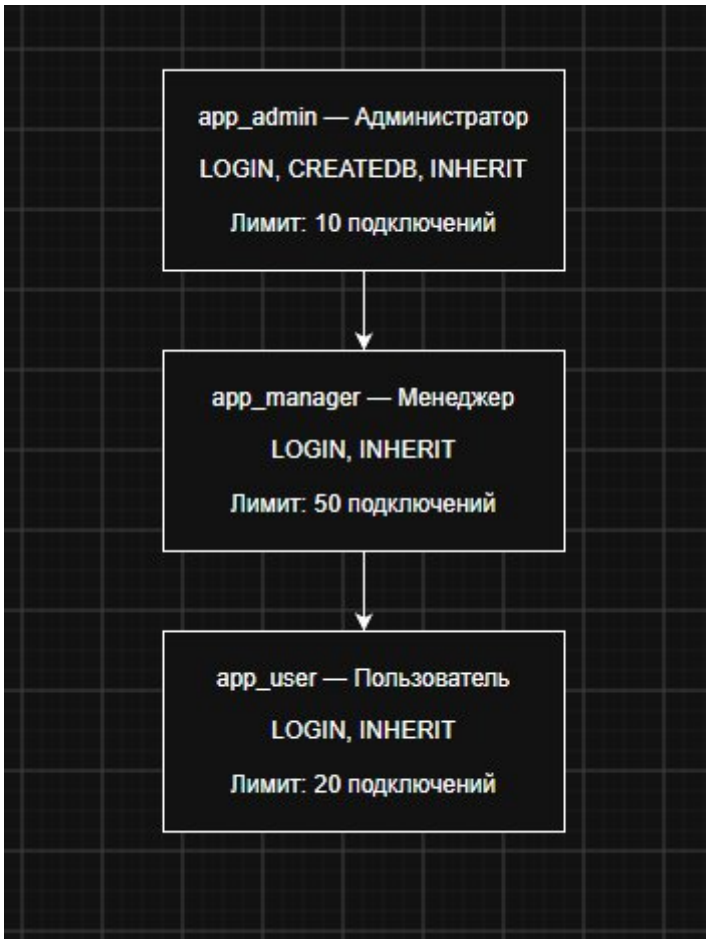
-- 6. Комментарии
INSERT INTO app.comments
(task_id, user_id, content, is_internal)
VALUES
(1, 1, 'Please make the design responsive', false),
(1, 2, 'Working on it', false),
(7, 1, 'Requirements approved', false),
(8, 2, 'Need clarification on database choice', true);
```

Добавлены данные

## 2.1

| **Объект / Операция** | **app_user** | **app_manager** | **app_admin** |
|---|---|---|---|
| app.users — SELECT | Только свои данные | Все пользователи | Все + системные поля |
| app.users — INSERT/UPDATE | Запрещено | Запрещено | Все операции |
| app.projects — SELECT | Свои проекты | Все проекты | Все проекты |
| app.projects — INSERT/UPDATE | Запрещено | Свои проекты | Все проекты |
| app.tasks — SELECT | Свои задачи | Все задачи в проектах | Все задачи |
| app.tasks — INSERT | Запрещено | В свои проекты | Все |
| app.tasks — UPDATE | Статус своих задач | Любые поля в своих проектах | Все |
| app.tasks — DELETE | Запрещено | Свои проекты | Все |
| app.comments — SELECT | Не-внутренние | Все, включая внутренние | Все |
| app.comments — INSERT/UPDATE | Свои комментарии | Все, включая is_internal | Все |
| app.task_history — SELECT | Запрещено | Свои проекты | Все |
| app.access_logs — SELECT | Запрещено | Запрещено | Все |
| DDL-операции | Запрещено | Запрещено | В своей схеме |

## 2.2


app_admin — Администратор
LOGIN, CREATEDB, INHERIT
Лимит: 10 подключений
app_manager — Менеджер
LOGIN, INHERIT
Лимит: 50 подключений
app_user — Пользователь
LOGIN, INHERIT
Лимит: 20 подключений

## 3.1

```sql
-- 1. Роль для подключения
CREATE ROLE app_connect NOLOGIN;

GRANT CONNECT ON DATABASE corporate_tasks TO app_connect;
GRANT USAGE ON SCHEMA app TO app_connect;
-- 2. Роль для чтения всех таблиц
CREATE ROLE app_read_all NOLOGIN;

GRANT app_connect TO app_read_all;
GRANT SELECT ON ALL TABLES IN SCHEMA app TO app_read_all;
-- 3. Роль для чтения справочников
CREATE ROLE app_read_reference NOLOGIN;

GRANT SELECT ON TABLE app.departments TO app_read_reference;
GRANT SELECT ON TABLE app.positions TO app_read_reference;
-- 4. Роль для работы с задачами
CREATE ROLE app_task_worker NOLOGIN;

GRANT app_read_reference TO app_task_worker;
GRANT SELECT ON TABLE app.tasks TO app_task_worker;
GRANT SELECT ON TABLE app.projects TO app_task_worker;
GRANT SELECT, INSERT ON TABLE app.comments TO app_task_worker;
GRANT USAGE ON SEQUENCE app.comments_comment_id_seq
TO app_task_worker;


-- 5. Роль для внутренних комментариев
CREATE ROLE app_internal_comments NOLOGIN;

GRANT SELECT (is_internal) ON TABLE app.comments
TO app_internal_comments;

GRANT UPDATE (is_internal) ON TABLE app.comments
TO app_internal_comments;

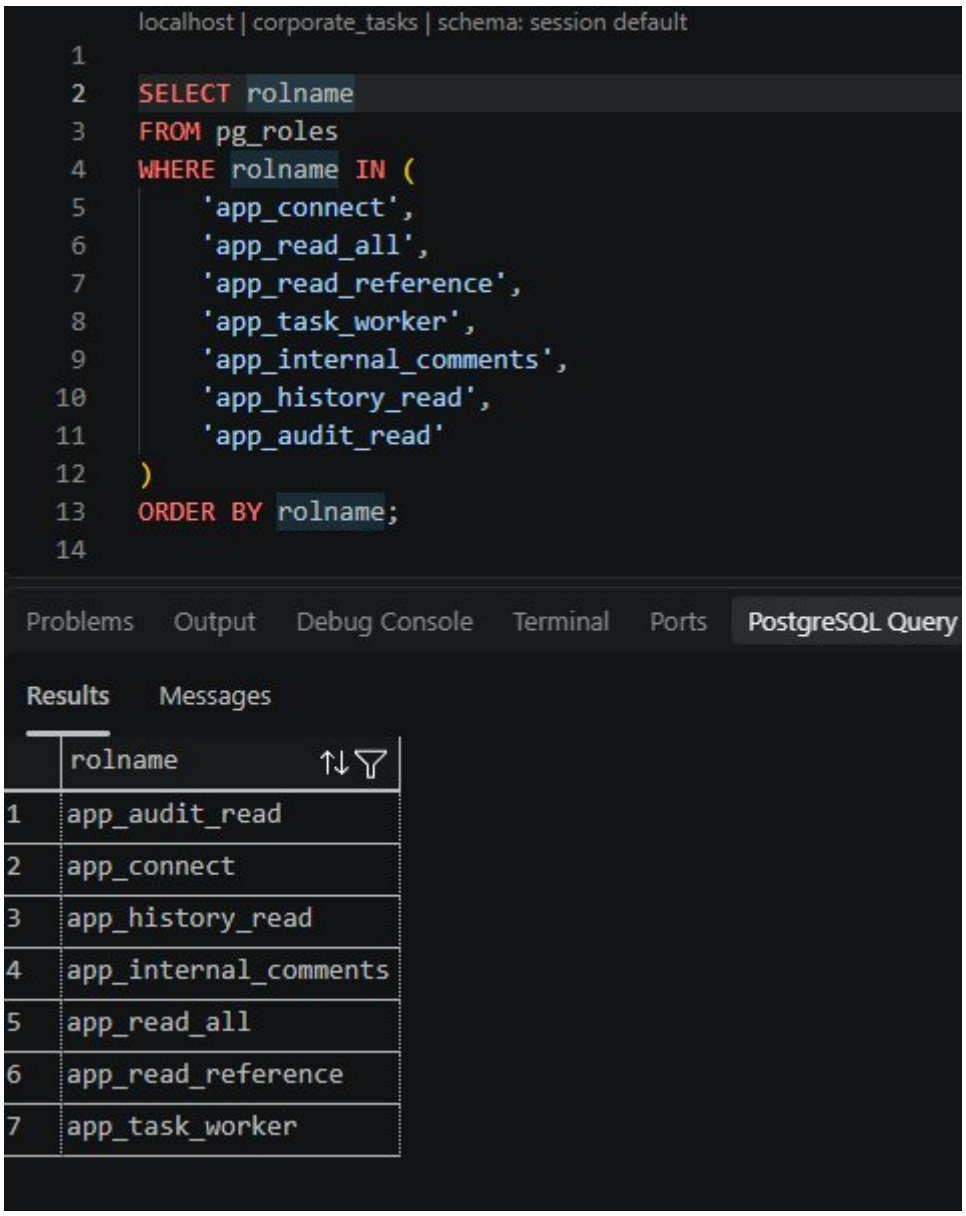

-- 6. Роль для чтения истории изменений
CREATE ROLE app_history_read NOLOGIN;

GRANT SELECT ON TABLE app.task_history TO app_history_read;


-- 7. Роль для чтения логов доступа
CREATE ROLE app_audit_read NOLOGIN;

GRANT SELECT ON TABLE app.access_logs TO app_audit_read;

```




Создание ролей и проверка создания этих ролей

```sql
-- Роль 1: Обычный пользователь

CREATE ROLE app_user WITH
    LOGIN
    PASSWORD 'UserPass123!'
    INHERIT
    CONNECTION LIMIT 20;

-- Назначение базовых прав
GRANT app_connect TO app_user;
GRANT app_read_reference TO app_user;
GRANT app_task_worker TO app_user;

-- Право читать таблицу пользователей
GRANT SELECT ON TABLE app.users TO app_user;
```

```sql
-- Роль 2: Менеджер

ALTER ROLE app_manager WITH
    LOGIN
    PASSWORD 'ManagerPass456!'
    INHERIT
    CONNECTION LIMIT 50;

-- Наследование прав пользователя
GRANT app_user TO app_manager;

-- Дополнительные роли
GRANT app_internal_comments TO app_manager;
GRANT app_history_read TO app_manager;

-- Права на проекты
GRANT SELECT ON ALL TABLES IN SCHEMA app TO app_manager;
GRANT INSERT, UPDATE ON TABLE app.projects TO app_manager;
GRANT USAGE ON SEQUENCE app.projects_project_id_seq
TO app_manager;

-- Права на задачи
GRANT SELECT, INSERT, UPDATE ON TABLE app.tasks
TO app_manager;
GRANT USAGE ON SEQUENCE app.tasks_task_id_seq
TO app_manager;

-- Права на комментарии
GRANT SELECT, INSERT, UPDATE, DELETE ON TABLE app.comments
TO app_manager;

-- Запись в историю изменений
GRANT INSERT ON TABLE app.task_history TO app_manager;
```

```sql
-- Роль 3: Администратор

ALTER ROLE app_admin WITH
    LOGIN
    PASSWORD 'AdminPass789!'
    CREATEDB
    INHERIT
    CONNECTION LIMIT 10;

-- Наследование прав менеджера
GRANT app_manager TO app_admin;

-- Полные права на таблицы и последовательности
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA app
TO app_admin;

GRANT ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA app
TO app_admin;

-- Право создавать объекты в схеме
GRANT CREATE ON SCHEMA app TO app_admin;

-- Доступ к журналу
GRANT app_audit_read TO app_admin;
GRANT SELECT, INSERT ON TABLE app.access_logs
TO app_admin;

GRANT USAGE ON SEQUENCE app.access_logs_log_id_seq
TO app_admin;

-- Управление пользователями
GRANT SELECT, INSERT, UPDATE, DELETE ON TABLE app.users
TO app_admin;
```

Создание пользователей и выдача прав

```sql
SELECT
    r.rolname AS "Роль",
    m.rolname AS "Полученные права роли"
FROM pg_roles r
JOIN pg_auth_members am ON r.oid = am.member
JOIN pg_roles m ON am.roleid = m.oid
WHERE r.rolname IN ('app_user', 'app_manager', 'app_admin')
ORDER BY r.rolname, m.rolname;
```

Problems Output Debug Console Terminal Ports **PostgreSQL Query Results**

Results Messages

| | Роль | Полученные права роли |
|---|---|---|
| 1 | app_admin | app_audit_read |
| 2 | app_admin | app_manager |
| 3 | app_manager | app_history_read |
| 4 | app_manager | app_internal_comments |
| 5 | app_manager | app_user |
| 6 | app_user | app_connect |
| 7 | app_user | app_read_reference |
| 8 | app_user | app_task_worker |

Проверка ролей и выданных им прав

## 3.3

```
326  -- 1. Обычный сотрудник
327  CREATE USER dev_alice WITH PASSWORD 'AliceDev123!';
328  GRANT app_user TO dev_alice;
329
330  -- 2. Менеджер проекта
331  CREATE USER pm_bob WITH PASSWORD 'BobPM456!';
332  GRANT app_manager TO pm_bob;
333
334  -- 3. Второй разработчик
335  CREATE USER dev_charlie WITH PASSWORD 'CharlieDev789!';
336  GRANT app_user TO dev_charlie;
337
338  -- 4. Администратор
339  CREATE USER admin_diana WITH PASSWORD 'DianaAdmin012!';
340  GRANT app_admin TO admin_diana;
341
342  -- 5. Маркетолог (только чтение)
343  CREATE USER marketing_eve WITH PASSWORD 'EveMarket345!';
344  GRANT app_read_all TO marketing_eve;
345
```

Создание сотрудников выдача ролей

```
3
4   SELECT
5       r.rolname AS "Пользователь",
6       ARRAY_AGG(m.rolname) AS "Роли"
7   FROM pg_roles r
8   JOIN pg_auth_members am ON r.oid = am.member
9   JOIN pg_roles m ON am.roleid = m.oid
10  WHERE r.rolname IN (
11      'dev_alice',
```

Problems Output Debug Console Terminal Ports PostgreSQL Query F

Results Messages

| | Пользователь | Роли |
|---|---|---|
| 1 | admin_diana | {app_admin} |
| 2 | dev_alice | {app_user} |
| 3 | dev_charlie | {app_user} |
| 4 | marketing_eve | {app_read_all} |
| 5 | pm_bob | {app_manager} |

Проверка назначенных ролей

```
3   ALTER TABLE app.users
4   ADD COLUMN db_role VARCHAR(63);
5
```

Problems Output Debug Console Terminal Ports PostgreSQL

Messages

```
9:04:55 PM     Started executing query at Line 2
               ALTER TABLE
               Total execution time: 00:00:00.008
```

Обновили пользователей и добавили колонку с ролями

localhost | corporate_tasks | schema: session default

```
1   UPDATE app.users
2   SET db_role = CASE username
3       WHEN 'alice' THEN 'dev_alice'
4       WHEN 'bob' THEN 'pm_bob'
5       WHEN 'charlie' THEN 'dev_charlie'
6       WHEN 'diana' THEN 'admin_diana'
7       WHEN 'eve' THEN 'marketing_eve'
8   END
9   WHERE username IN (
10      'alice', 'bob', 'charlie', 'diana', 'eve'
11  );
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages

```
9:08:56 PM     Started executing query at Line 1
               UPDATE 5
               Total execution time: 00:00:00.003
```

Связываем сотрудников с учетными записями

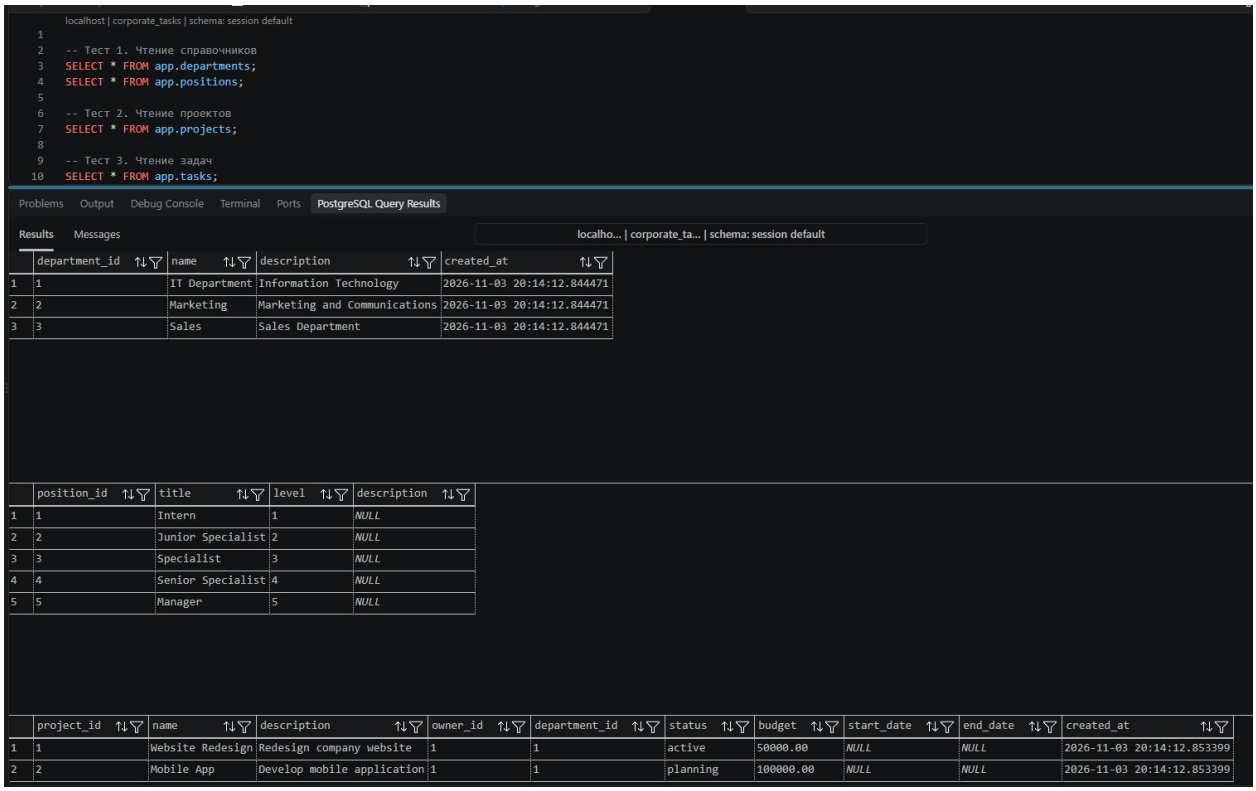
```
4   SELECT username, db_role
5   FROM app.users
6   ORDER BY username;
7
```

Problems Output Debug Console Terminal Por

Results Messages

| | username | db_role |
|---|---|---|
| 1 | alice | dev_alice |
| 2 | bob | pm_bob |
| 3 | charlie | dev_charlie |
| 4 | diana | admin_diana |
| 5 | eve | marketing_eve |

# 4.1 app.user(dev_alice)



```
localhost | corporate_tasks | schema: session default

-- Тест 1. Чтение справочников
SELECT * FROM app.departments;
SELECT * FROM app.positions;

-- Тест 2. Чтение проектов
SELECT * FROM app.projects;

-- Тест 3. Чтение задач
SELECT * FROM app.tasks;
```

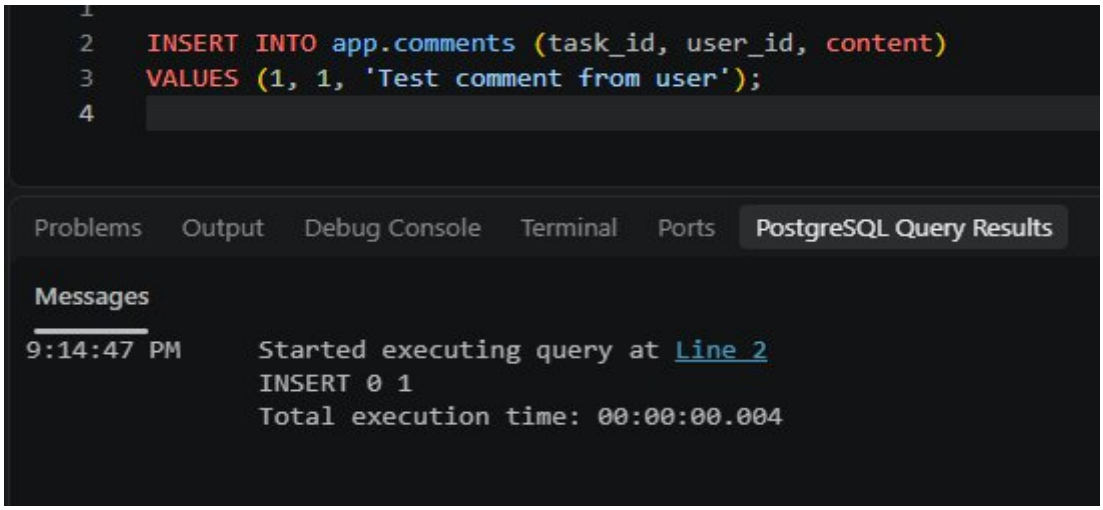
Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... | corporate_ta... | schema: session default

| | department_id | name | description | created_at |
|---|---|---|---|---|
| 1 | 1 | IT Department | Information Technology | 2026-11-03 20:14:12.844471 |
| 2 | 2 | Marketing | Marketing and Communications | 2026-11-03 20:14:12.844471 |
| 3 | 3 | Sales | Sales Department | 2026-11-03 20:14:12.844471 |

| | position_id | title | level | description |
|---|---|---|---|---|
| 1 | 1 | Intern | 1 | NULL |
| 2 | 2 | Junior Specialist | 2 | NULL |
| 3 | 3 | Specialist | 3 | NULL |
| 4 | 4 | Senior Specialist | 4 | NULL |
| 5 | 5 | Manager | 5 | NULL |

| | project_id | name | description | owner_id | department_id | status | budget | start_date | end_date | created_at |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Website Redesign | Redesign company website | 1 | 1 | active | 50000.00 | NULL | NULL | 2026-11-03 20:14:12.853399 |
| 2 | 2 | Mobile App | Develop mobile application | 1 | 1 | planning | 100000.00 | NULL | NULL | 2026-11-03 20:14:12.853399 |

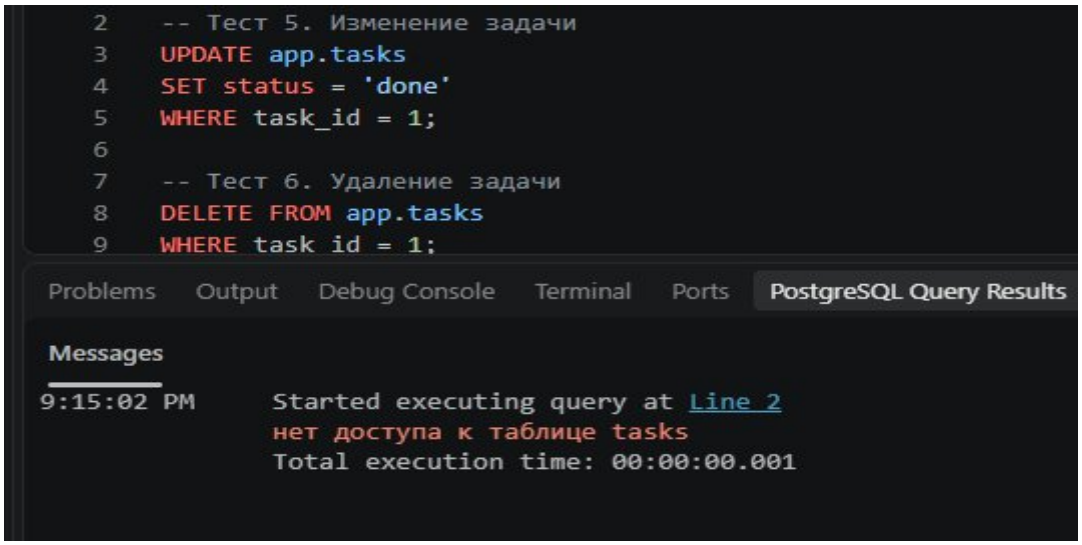


```
INSERT INTO app.comments (task_id, user_id, content)
VALUES (1, 1, 'Test comment from user');
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages

```
9:14:47 PM    Started executing query at Line 2
              INSERT 0 1
              Total execution time: 00:00:00.004
```



```
-- Тест 5. Изменение задачи
UPDATE app.tasks
SET status = 'done'
WHERE task_id = 1;

-- Тест 6. Удаление задачи
DELETE FROM app.tasks
WHERE task_id = 1;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages

```
9:15:02 PM    Started executing query at Line 2
              нет доступа к таблице tasks
              Total execution time: 00:00:00.001
```

```sql
SELECT user_id, username, email, full_name
FROM app.users;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

| | user_id | username | email | full_name |
|---|---|---|---|---|
| 1 | 1 | alice | alice@company.com | Alice Johnson |
| 2 | 2 | bob | bob@company.com | Bob Smith |
| 3 | 3 | charlie | charlie@company.com | Charlie Brown |
| 4 | 4 | diana | diana@company.com | Diana Prince |
| 5 | 5 | eve | eve@company.com | Eve Wilson |

```sql
SELECT * FROM app.access_logs;
```

Problems Output Debug Console Terminal Ports PostgreS

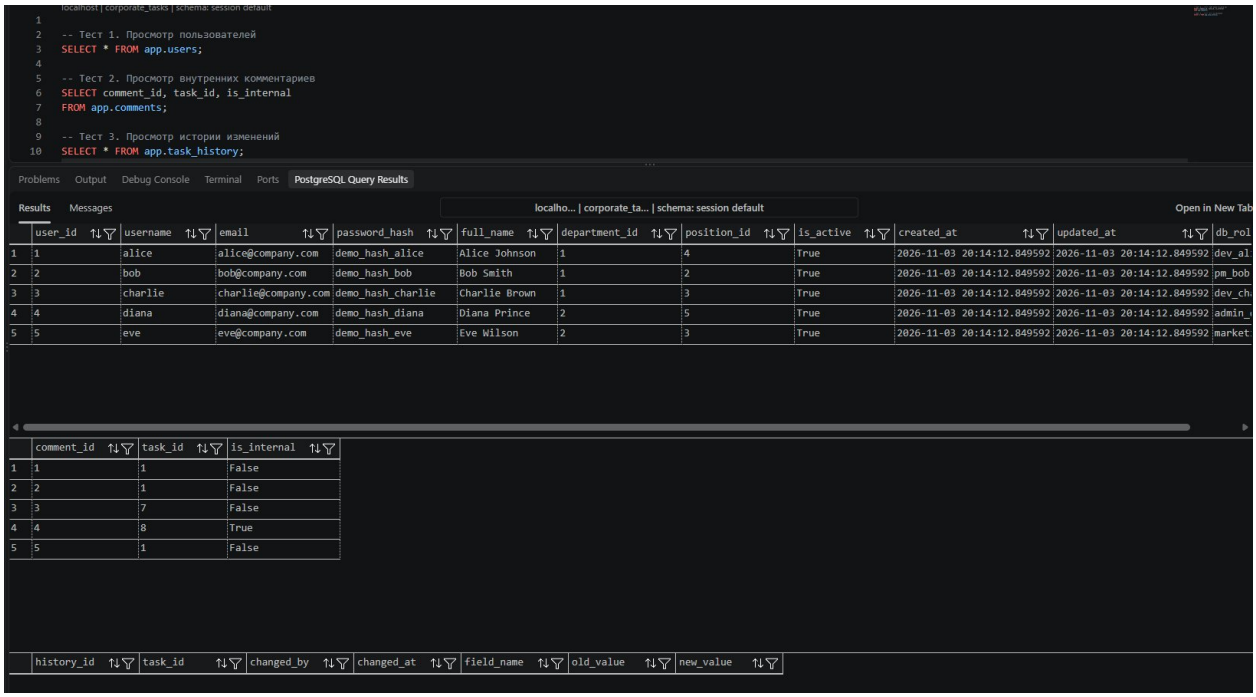
Messages

9:15:46 PM Started executing query at Line 2

нет доступа к таблице access_logs

Total execution time: 00:00:00.001

# 4.2 app_manager (pm_bob)



```
-- Тест 1. Просмотр пользователей
SELECT * FROM app.users;

-- Тест 2. Просмотр внутренних комментариев
SELECT comment_id, task_id, is_internal
FROM app.comments;

-- Тест 3. Просмотр истории изменений
SELECT * FROM app.task_history;
```

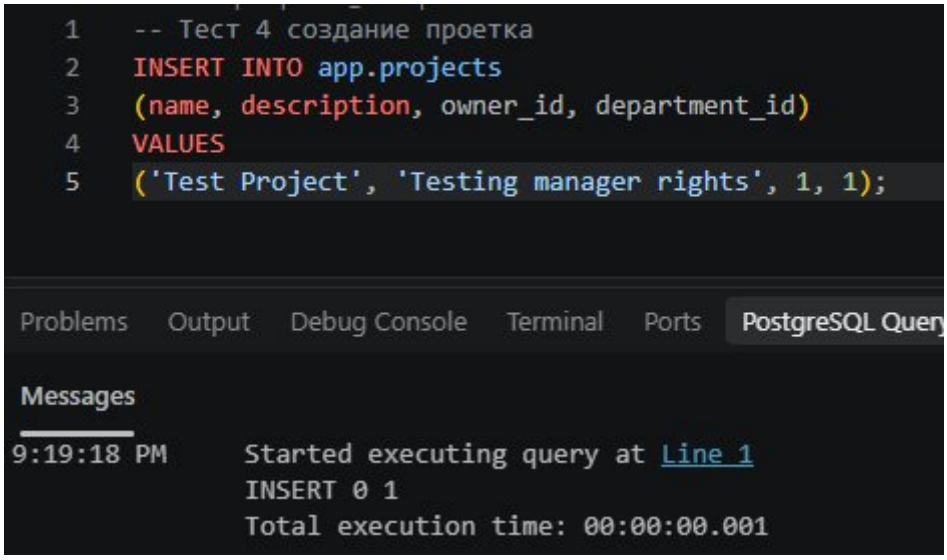
Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

| | user_id | username | email | password_hash | full_name | department_id | position_id | is_active | created_at | updated_at | db_rol |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | alice | alice@company.com | demo_hash_alice | Alice Johnson | 1 | 4 | True | 2026-11-03 20:14:12.849592 | 2026-11-03 20:14:12.849592 | dev_al |
| 2 | 2 | bob | bob@company.com | demo_hash_bob | Bob Smith | 1 | 2 | True | 2026-11-03 20:14:12.849592 | 2026-11-03 20:14:12.849592 | pm_bob |
| 3 | 3 | charlie | charlie@company.com | demo_hash_charlie | Charlie Brown | 1 | 3 | True | 2026-11-03 20:14:12.849592 | 2026-11-03 20:14:12.849592 | dev_ch |
| 4 | 4 | diana | diana@company.com | demo_hash_diana | Diana Prince | 2 | 5 | True | 2026-11-03 20:14:12.849592 | 2026-11-03 20:14:12.849592 | admin_ |
| 5 | 5 | eve | eve@company.com | demo_hash_eve | Eve Wilson | 2 | 3 | True | 2026-11-03 20:14:12.849592 | 2026-11-03 20:14:12.849592 | market |

| | comment_id | task_id | is_internal |
|---|---|---|---|
| 1 | 1 | 1 | False |
| 2 | 2 | 1 | False |
| 3 | 3 | 7 | False |
| 4 | 4 | 8 | True |
| 5 | 5 | 1 | False |

| history_id | task_id | changed_by | changed_at | field_name | old_value | new_value |
|---|---|---|---|---|---|---|

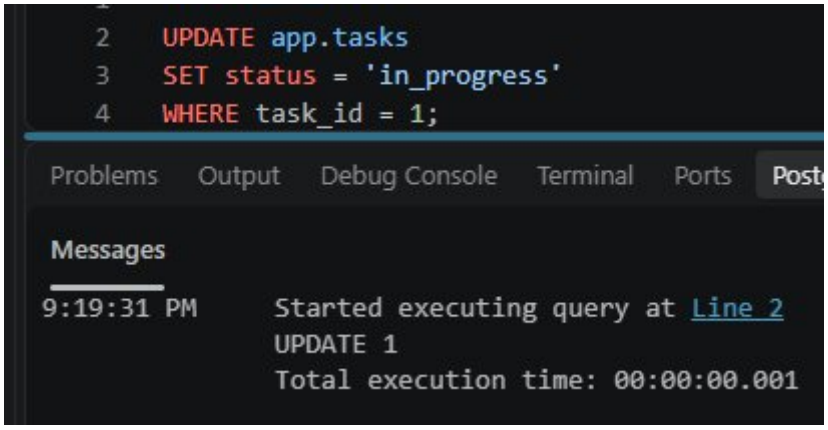


```
-- Тест 4 создание проетка
INSERT INTO app.projects
(name, description, owner_id, department_id)
VALUES
('Test Project', 'Testing manager rights', 1, 1);
```

Problems Output Debug Console Terminal Ports PostgreSQL Query

Messages

```
9:19:18 PM    Started executing query at Line 1
              INSERT 0 1
              Total execution time: 00:00:00.001
```



```
UPDATE app.tasks
SET status = 'in_progress'
WHERE task_id = 1;
```

Problems Output Debug Console Terminal Ports Post

Messages

```
9:19:31 PM    Started executing query at Line 2
              UPDATE 1
              Total execution time: 00:00:00.001
```

```
2   DELETE FROM app.tasks
3   WHERE task_id = 1;
```

Problems Output Debug Console Terminal Ports Pos

**Messages**

```
9:19:54 PM      Started executing query at Line 2
                нет доступа к таблице tasks
                Total execution time: 00:00:00
```

```
2   INSERT INTO app.task_history
3   (task_id, changed_by, field_name, old_value, new_value)
4   VALUES
5   (1, 1, 'status', 'todo', 'in_progress');
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

**Messages**

```
9:20:11 PM      Started executing query at Line 2
                нет доступа к последовательности task_history_history_id_seq
                Total execution time: 00:00:00
```

```
2   SELECT * FROM app.access_logs;
3
```

roblems Output Debug Console Terminal Ports PostgreSQL Query Results

esults Messages localho... | corporate_ta... | schema: session default

| log_id | user_id | action | resource_... | resource_... | ip_address | logged_at |
|---|---|---|---|---|---|---|

```
3   INSERT INTO app.tasks (project_id, title)
4   SELECT project_id, 'Manager test task'
5   FROM app.projects
6   WHERE name = 'Test Project'
7   ORDER BY project_id DESC
8   LIMIT 1;
```

Problems Output Debug Console Terminal Ports PostgreSQL C

**Messages**

```
9:21:17 PM      Started executing query at Line 2
                INSERT 0 1
                Total execution time: 00:00:00.002
```

\

## 4.3 app_admin (admin_diana)

```
2   -- Тест 1. Просмотр журнала доступа
3   SELECT * FROM app.access_logs;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... | corporate_ta... | schema: session default

| log_id | user_id | action | resource_... | resource_... | ip_address | logged_at |
|---|---|---|---|---|---|---|

```
2   -- Тест 2. Изменение данных пользователя
3   UPDATE app.users
4   SET is_active = false
5   WHERE username = 'bob';
```

Problems Output Debug Console Terminal Ports Postgre

Messages

```
:24:57 PM     Started executing query at Line 2
              UPDATE 1
              Total execution time: 00:00:00.004
```

```
2   -- Тест 3. Создание таблицы
3   CREATE TABLE app.test_admin (
4       id INT,
5       name TEXT
6   );
7
8   -- Тест 4. Добавление нового столбца
9   ALTER TABLE app.test_admin
10  ADD COLUMN created_at TIMESTAMP;
11
12  -- Тест 5. Удаление тестовой таблицы
13  DROP TABLE app.test_admin;
```

roblems Output Debug Console Terminal Ports PostgreSQL Query Result

essages

```
25:15 PM      Started executing query at Line 2
              CREATE TABLE
25:15 PM      Started executing query at Line 8
              ALTER TABLE
25:15 PM      Started executing query at Line 12
              DROP TABLE
              Total execution time: 00:00:00.036
```

```sql
-- Тест 6. Просмотр ролей
SELECT rolname, rolsuper, rolcreatedb
FROM pg_roles
WHERE rolname LIKE 'app_%';
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Resul

Results Messages

| | rolname | rolsuper | rolcreatedb |
|---|---|---|---|
| 1 | app_guest | False | False |
| 2 | app_employee | False | False |
| 3 | app_superuser | False | False |
| 4 | app_connect | False | False |
| 5 | app_read_all | False | False |
| 6 | app_read_reference | False | False |
| 7 | app_task_worker | False | False |
| 8 | app_internal_comments | False | False |
| 9 | app_history_read | False | False |
| 10 | app_audit_read | False | False |
| 11 | app_user | False | False |
| 12 | app_manager | False | False |
| 13 | app_admin | False | True |

```sql
-- Тест 7. Просмотр прав ролей на таблицы

SELECT
    grantee,
    table_name,
    privilege_type
FROM information_schema.role_table_grants
WHERE grantee IN (
    'app_user',
    'app_manager',
    'app_admin'
);
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

| | grantee | table_name | privilege_type |
|---|---|---|---|
| 1 | app_manager | tasks | INSERT |
| 2 | app_manager | tasks | SELECT |
| 3 | app_manager | tasks | UPDATE |
| 4 | app_admin | tasks | INSERT |
| 5 | app_admin | tasks | SELECT |
| 6 | app_admin | tasks | UPDATE |
| 7 | app_admin | tasks | DELETE |
| 8 | app_admin | tasks | TRUNCATE |
| 9 | app_admin | tasks | REFERENCES |
| 10 | app_admin | tasks | TRIGGER |
| 11 | app_manager | access_logs | SELECT |
| 12 | app_admin | access_logs | INSERT |
| 13 | app_admin | access_logs | SELECT |
| 14 | app_admin | access_logs | UPDATE |
| 15 | app_admin | access_logs | DELETE |
| 16 | app_admin | access_logs | TRUNCATE |
| 17 | app_admin | access_logs | REFERENCES |
| 18 | app_admin | access_logs | TRIGGER |
| 19 | app_user | users | SELECT |
| 20 | app_manager | users | SELECT |
| 21 | app_admin | users | INSERT |

# 5.1

```sql
-- 5.1. Список всех ролей с атрибутами
SELECT
    rolname AS "Роль",
    rolsuper AS "Superuser",
    rolcreatedb AS "Create DB",
    rolcreaterole AS "Create Role",
    rolcanlogin AS "Login",
    rolinherit AS "Inherit",
    rolconnlimit AS "Conn Limit",
    rolvaliduntil AS "Valid Until"
FROM pg_roles
WHERE rolname LIKE 'app_%'
   OR rolname IN (
        'dev_alice',
        'pm_bob',
        'dev_charlie',
        'admin_diana',
        'marketing_eve'
   )
ORDER BY rolname;
```

| | Роль | Superuser | Create DB | Create Role | Login | Inherit | Conn Limit | Valid Until |
|---|---|---|---|---|---|---|---|---|
| 1 | admin_diana | False | False | False | True | True | -1 | *NULL* |
| 2 | app_admin | False | True | False | True | True | 10 | *NULL* |
| 3 | app_audit_read | False | False | False | False | True | -1 | *NULL* |
| 4 | app_connect | False | False | False | False | True | -1 | *NULL* |
| 5 | app_employee | False | False | False | False | True | -1 | *NULL* |
| 6 | app_guest | False | False | False | False | True | -1 | *NULL* |
| 7 | app_history_read | False | False | False | False | True | -1 | *NULL* |
| 8 | app_internal_comments | False | False | False | False | True | -1 | *NULL* |
| 9 | app_manager | False | False | False | True | True | 50 | *NULL* |
| 10 | app_read_all | False | False | False | False | True | -1 | *NULL* |
| 11 | app_read_reference | False | False | False | False | True | -1 | *NULL* |
| 12 | app_superuser | False | False | False | False | True | -1 | *NULL* |
| 13 | app_task_worker | False | False | False | False | True | -1 | *NULL* |
| 14 | app_user | False | False | False | True | True | 20 | *NULL* |
| 15 | dev_alice | False | False | False | True | True | -1 | *NULL* |
| 16 | dev_charlie | False | False | False | True | True | -1 | *NULL* |
| 17 | marketing_eve | False | False | False | True | True | -1 | *NULL* |
| 18 | pm_bob | False | False | False | True | True | -1 | *NULL* |

```sql
-- 5.1. Иерархия ролей

SELECT
    r.rolname AS "Роль",
    m.rolname AS "Наследует от",
    a.admin_option AS "С правом передачи"
FROM pg_roles r
JOIN pg_auth_members am
    ON r.oid = am.member
JOIN pg_roles m
    ON am.roleid = m.oid
JOIN pg_auth_members a
    ON a.member = r.oid
   AND a.roleid = m.oid
WHERE r.rolname IN (
    'dev_alice',
    'pm_bob',
    'dev_charlie',
    'admin_diana',
    'marketing_eve'
)
ORDER BY r.rolname, m.rolname;

```

| | Роль | Наследует от | С правом передачи |
|---|---|---|---|
| 1 | admin_diana | app_admin | False |
| 2 | dev_alice | app_user | False |
| 3 | dev_charlie | app_user | False |
| 4 | marketing_eve | app_read_all | False |
| 5 | pm_bob | app_manager | False |

```sql
-- 5.1. Права на таблицы

SELECT
    grantee AS "Роль",
    table_schema AS "Схема",
    table_name AS "Таблица",
    STRING_AGG(privilege_type, ', ') AS "Привилегии"
FROM information_schema.role_table_grants
WHERE grantee LIKE 'app_%'
  AND table_schema = 'app'
GROUP BY grantee, table_schema, table_name
ORDER BY grantee, table_name;

```

| | Роль | Схема | Таблица | Привилегии |
|---|---|---|---|---|
| 1 | app_admin | app | access_logs | INSERT, TRIGGER, REFERENCES, TRUNCATE, DELETE, UPD… |
| 2 | app_admin | app | comments | INSERT, TRIGGER, REFERENCES, TRUNCATE, DELETE, UPD… |
| 3 | app_admin | app | departments | INSERT, TRIGGER, REFERENCES, TRUNCATE, DELETE, UPD… |
| 4 | app_admin | app | positions | SELECT, INSERT, UPDATE, DELETE, TRUNCATE, REFERENC… |
| 5 | app_admin | app | projects | DELETE, TRUNCATE, REFERENCES, INSERT, TRIGGER, SEL… |
| 6 | app_admin | app | task_history | TRUNCATE, TRIGGER, INSERT, SELECT, UPDATE, DELETE,… |
| 7 | app_admin | app | tasks | INSERT, DELETE, TRUNCATE, UPDATE, SELECT, TRIGGER,… |
| 8 | app_admin | app | users | DELETE, INSERT, SELECT, UPDATE, TRUNCATE, REFERENC… |
| 9 | app_audit_read | app | access_logs | SELECT |
| 10 | app_history_read | app | task_history | SELECT |
| 11 | app_manager | app | access_logs | SELECT |
| 12 | app_manager | app | comments | INSERT, UPDATE, DELETE, SELECT |
| 13 | app_manager | app | departments | SELECT |
| 14 | app_manager | app | positions | SELECT |
| 15 | app_manager | app | projects | SELECT, INSERT, UPDATE |
| 16 | app_manager | app | task_history | SELECT, INSERT |
| 17 | app_manager | app | tasks | INSERT, SELECT, UPDATE |
| 18 | app_manager | app | users | SELECT |
| 19 | app_read_all | app | access_logs | SELECT |
| 20 | app_read_all | app | comments | SELECT |
| 21 | app_read_all | app | departments | SELECT |
| 22 | app_read_all | app | positions | SELECT |
| 23 | app_read_all | app | projects | SELECT |
| 24 | app_read_all | app | task_history | SELECT |
| 25 | app_read_all | app | tasks | SELECT |
| 26 | app_read_all | app | users | SELECT |
| 27 | app_read_reference | app | departments | SELECT |
| 28 | app_read_reference | app | positions | SELECT |
| 29 | app_task_worker | app | comments | SELECT, INSERT |
| 30 | app_task_worker | app | projects | SELECT |
| 31 | app_task_worker | app | tasks | SELECT |
| 32 | app_user | app | users | SELECT |

```sql
-- 5.1. Активные подключения

SELECT
    usename AS "Пользователь",
    datname AS "База данных",
    client_addr AS "IP",
    backend_start AS "Начало сессии",
    state AS "Состояние"
FROM pg_stat_activity
WHERE datname = 'corporate_tasks';
```

Problems Output Debug Console Terminal Ports **PostgreSQL Query Results**

Results Messages localho... | corp

| | Пользователь | База данных | IP | Начало сессии | Состояние |
|---|---|---|---|---|---|
| 1 | postgres | corporate_tasks | ::1 | 2026-11-03 20:07:07.069904+05 | idle |
| 2 | postgres | corporate_tasks | ::1 | 2026-11-03 20:07:07.215728+05 | idle |
| 3 | dev_alice | corporate_tasks | ::1 | 2026-11-03 21:13:54.946706+05 | idle |
| 4 | pm_bob | corporate_tasks | ::1 | 2026-11-03 21:18:02.694876+05 | idle |
| 5 | admin_diana | corporate_tasks | ::1 | 2026-11-03 21:24:19.743142+05 | idle |
| 6 | postgres | corporate_tasks | ::1 | 2026-11-03 21:32:56.243992+05 | active |
| 7 | postgres | corporate_tasks | ::1 | 2026-11-03 21:34:15.795215+05 | idle |